# Should we teach using schemas? Evidence from a randomised trial

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**CONTEXT** Schema-based instruction may alter knowledge organisation and diagnostic reasoning strategies through the provision of structured knowledge to novice trainees. The effects of schema-based instruction on diagnostic accuracy and knowledge organisation have not been rigorously tested.

**METHODS** Year 2 medical students were randomised to learn four cardiac diagnoses using schema-based instruction ($n$ = 26) or traditional instruction ($n$ = 27) on a high-fidelity cardiopulmonary simulator (CPS). Students completed case-based learning in groups of two to five and underwent individual written and practical tests. The written test consisted of questions testing features that linked or distinguished diagnoses (structured knowledge) and questions testing features of individual diagnoses (factual knowledge). A practical test of diagnostic accuracy on the CPS was performed for two diagnoses present in the learning phase (taught lesions) and two untaught lesions. A majority of students ($n$ = 37, 70%) voluntarily returned for follow-up written testing 2–4 weeks later.

**RESULTS** Learning time and accuracy did not differ between students on schema-based and those on traditional instruction. Students receiving schema-based instruction performed better on structured knowledge questions ($p < 0.001$) and no differently on factual knowledge questions ($p = 0.7$). Relative differences between groups remained unchanged on follow-up testing. Diagnostic success was higher in the schema-based instruction group for taught lesions (mean difference = 38%, 95% confidence interval [CI] 20–56; $p < 0.001$) and untaught lesions (mean difference = 31%, 95% CI 15–48; $p < 0.001$).

**CONCLUSIONS** Schema-based instruction was associated with improved retention of structured knowledge and diagnostic performance among novices. This study provides important proof-of-concept for a schema-based approach and suggests there is substantial benefit to using this approach with novice trainees.

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## INTRODUCTION

The amount and complexity of information medical students are asked to master are exponentially increasing. One approach to help students overcome this information overload is to identify strategies that help students acquire knowledge more efficiently. Cognitive load theory and knowledge organisation concepts can help in designing such learning strategies.

### Information overload through the lens of cognitive load theory

Cognitive load theory views the challenges of learning through the finite limitations of working memory.[1] Working memory can process only three to seven information elements.[1] When a subject matter involves multiple interacting elements, it can overwhelm working memory and become difficult to learn. Traditionally, complex subjects with many interacting elements are taught by presenting each element separately to reduce cognitive load.[1,2] For example, learning cardiac physical examination requires students to master lists of findings for each relevant diagnosis. However, when students face a patient with an unknown systolic murmur, they must integrate all of this information to arrive at a diagnosis, a task that presents a tremendous challenge to working memory. An alternative approach is to explicitly teach how the elements interact by organising information into schemas[1] or representations of diagnoses distinguished by relevant clinical and pathophysiologic variables. Schemas reduce cognitive load by presenting the most important links between interacting elements.

### Schemas and knowledge organisation

Items of knowledge acquired by novices are often stored in long-term memory as isolated facts. Isolated facts are hard to recall and apply to clinical problems because they are not linked to other relevant information.[3] In order for this information to be useful in solving clinical problems, it needs to be reorganised around clinical findings and concepts.[2] This requires trainees to link multiple causal hypotheses and elaborate on them in order to build an understanding of the underlying concept. Bordage describes this highly linked conceptual knowledge as 'compiled'.[3] Schemas group diagnoses by relevant clinical and pathophysiologic variables. Therefore, the provision of schemas is hypothesised to aid in the development of compiled knowledge because schemas present the information in a structured, clinically relevant way.

### Schemas and clinical reasoning

Diagnostic reasoning strategies used by novice learners include hypothetico-deductive and schema-based reasoning. Elstein *et al.*[4] describe hypothetico-deductive reasoning as a process whereby clinicians formulate and test hypotheses in a sequential fashion. By contrast, schema-based reasoning is a process in which key clinical features are used to include or exclude sets of diagnoses. In an observational study, Coderre *et al.*[5] found that novice medical students achieved higher diagnostic success when employing a schema-based strategy compared with a hypothetico-deductive strategy. It is tempting to attribute the higher diagnostic success to schema-based reasoning. However, the use of schemas may also be a surrogate for students with a more thorough understanding of the clinical problem.[5]

### Relating knowledge organisation and reasoning strategies

Knowledge organisation and reasoning strategies are likely to be linked.[3,6] Dispersed knowledge acquired through rote memorisation might force hypothetico-deductive reasoning. Conversely, highly compiled knowledge stored as elaborate causal networks may promote schema-based reasoning. Therefore, teaching schemas may have beneficial effects in both promoting more expert knowledge organisation and facilitating expert-like reasoning.

### Testing a schema-based approach

Previous studies of schema-based instruction identified improved test scores and better diagnostic accuracy when schemas were compared with traditional lists or tables for teaching in nephrology[7,8] and congenital cardiology.[9] These studies suggest that schemas provide information in a structure that enables the use of schema-based reasoning. However, this evidence comes from non-randomised retrospective comparisons in which the effect of schemas may have been confounded by differences among students in level of expertise.[7–9]

Seasoned clinicians have been critical of schema-based instruction, arguing that schemas cannot prepare students to reason through an unlearned diagnosis.[8,9] Even experts abandon schematic reasoning and adopt hypothesis testing when faced with an unfamiliar problem.[10] Novice students taught with a schema have not had to navigate the material using a hypothetico-deductive

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approach. As a result, students may struggle when encountering untaught diagnoses.

Cardiac physical diagnosis provides a good model to test the effect of schema-based instructional guidance for several reasons. Firstly, multiple studies have documented poor cardiac auscultatory skills among senior medical students and residents, and resulting low rates of diagnostic success (19–55%).[11–15] Secondly, traditional teaching of cardiac physical diagnosis involves diagnosis-specific lists of features, structured around steps in the cardiac physical examination. This approach leads a novice learner to hypothetico-deductive reasoning until repeated encounters promote the development of individual schemas. Thirdly, a high-fidelity cardiopulmonary simulator (CPS), Harvey® (Gorden Center for Research in Medical Education, University of Miami, Miami, FL, USA), which can mimic a wide range of clinically relevant pathology for learning and testing,[16] is available.

We aimed to test the effects of schema-based instruction in comparison with those of a traditional framework on the acquisition of cardiac auscultation skills among novices in a randomised trial. We hypothesised that schema-based instruction would foster clinically relevant knowledge (hereafter referred to as 'structured knowledge'), increase schema-based reasoning and improve diagnostic accuracy. We also hypothesised that schema-based instruction would increase diagnostic error compared with traditional instruction for lesions that students had not encountered before.

## METHODS

### Population

Year 2 medical students at the University of Toronto were invited to participate. Students had previously completed two 4-hour sessions covering the components of the cardiac physical examination. All participants provided informed consent.

### Design

The study design featured case-based learning using Harvey®, a CPS that reproduces visual, tactile and auditory findings of the cardiac examination, designed at the Gordon Center for Research in Medical Education, University of Miami (Miami, FL, USA). Participants attended a 15-minute orientation.

Participants attended one of 14 sessions, each of which included up to five participants. Sessions were randomised into two categories to include a control sample that received an instructional framework based on the traditional approach (Fig. 1) and an intervention sample that received an instructional framework based on a diagnostic schema (Fig. 2). Both frameworks contained the same information. However, the organisation of the information differed. The traditional framework (Fig. 1) was structured around individual valvular lesions and listed the 10 clinical features of each diagnosis in a table format. The schema framework (Fig. 2) grouped diagnoses according to clinical features based on murmur location and timing (e.g. 'apex beat', 'radiation' and 'character' for a murmur heard at the apex).

Before the schemas were developed, we conducted a search for existing frameworks in the literature. This revealed two schemas that were inappropriate for the study. One schema used a finding not available on the simulator, hepatomegaly, as a key distinguishing feature of cardiac lesions,[17] whereas the other was a cumbersome, long framework beyond the scope of a novice.[18] Therefore, two experts with 10 years of experience in cardiac physical examination were consulted to develop an appropriate schema. The schema was revised based on pilot data.

The study was divided into four phases: learning; consolidation; testing, and retesting. The first three phases all occurred in the same session. Retesting occurred 2–4 weeks later. All written materials were handed to the facilitator at the end of the session to prevent any contamination between groups.

### Learning phase

In groups of two to five students, participants were asked to deduce the diagnosis of four unknown cardiac lesions (aortic regurgitation, aortic stenosis, mitral regurgitation and ventricular septal defect) on the simulator using the assigned framework. A facilitator was available for technical troubleshooting, but did not participate in the learning process. Each participant was asked to record his or her diagnosis and the key features that had led to it. Once the group had finished, participants were given a written answer guide. The group was allowed 15 minutes to resolve any incorrect diagnoses or clarify examination features. This case-based, collaborative design was intended to foster learning while limiting the effects of instructional guidance to the framework provided. The four lesions chosen were derived from a survey of

| | Aortic stenosis | Aortic sclerosis | Mitral regurgitation | Mitral Valve prolapse | Ventricular septal defect | Hypertrophic cardiomyopathy | Tricuspid regurgitation | Aortic regurgitation | Mitral stenosis | Pericardial rub |
|---|---|---|---|---|---|---|---|---|---|---|
| Timing | Systolic | Systolic | Systolic | Systolic | Systolic | Systolic | Systolic | Diastolic | Diastolic | Systolic & diastolic |
| Shape | | | | | | | | | | |
| Location | Base | Base | Apex | Apex | Border | Base & apex | Border | Border | Apex | Border |
| Radiation | Carotid | None | Axilla | None | None | Base & apex | None | None | None | None |
| Pitch | Harsh, course | N/A | Blowing | N/A | Harsh | Blowing | Blowing | High pitch, hollow | Low pitch, rumble | Musical |
| Carotid upstroke | Decreased | Normal | Normal | Normal | Normal | Double pulsation | Normal | Bounding | Difficult to palpate | Normal |
| JVP | Normal | Normal | Normal | Normal | Normal | Normal | CV wave | Normal | Normal | Normal |
| Palpation | Sustained apex | Normal | Displaced apex | Normal | Parasternal thrill | Triple apex | RV heave | Displaced apex | Difficult to palpate | Normal |
| Extra heart sounds | +/- S4 | None | +S3 | Mid-systolic click | Loud P2 | +/- S3 or S4 | None | +/- S3 | Opening snap | None |
| Inspiration | No change | No change | No change | No change | No change | No change | Increases | No change | No change | No change |

**Figure 1** Traditional instructional framework

internists who identified them as key components of the curriculum for general practice.[12] Learning worksheets were reviewed to ensure no misconceptions were learned using either framework. Each diagnosis was scored as correct or incorrect. Key features were categorised as correct features in the schema, correct features not in the schema, or incorrect features.

### Consolidation phase

Participants were allocated 15 minutes to review the instructional framework prior to testing. They were allowed to discuss any learning difficulties within their group.

### Testing and retesting phase

Participants individually completed a written test and then a practical test without the aid of their instructional framework. Participants were invited to undertake a repeat written test administered 2–4 weeks after the initial session. Retesting was limited to the written test by logistical constraints.

### Written test

Participants completed a 19-question written test including single-response multiple-choice questions (MCQs), multiple-response MCQs and short-answer questions. The questions were divided into two groups. The first group ($n$ = 9) tested features which linked or distinguished diagnoses (structured knowledge questions). The second group ($n$ = 10) tested features of individual diagnoses (factual knowledge questions). Sample questions from each group are included in Table 1. Both groups involved diagnoses familiar to the participants from the learning phase (taught lesions) and not taught to the participants in the learning phase (untaught lesions). All tested diagnoses were present in both the structured and factual knowledge questions. The written test was marked by two researchers blinded to group allocation. Disagreement was resolved by consensus.

### Practical test

Participants were asked to individually identify four unidentified lesions on Harvey®, recording the diagnosis and key features used in arriving at the

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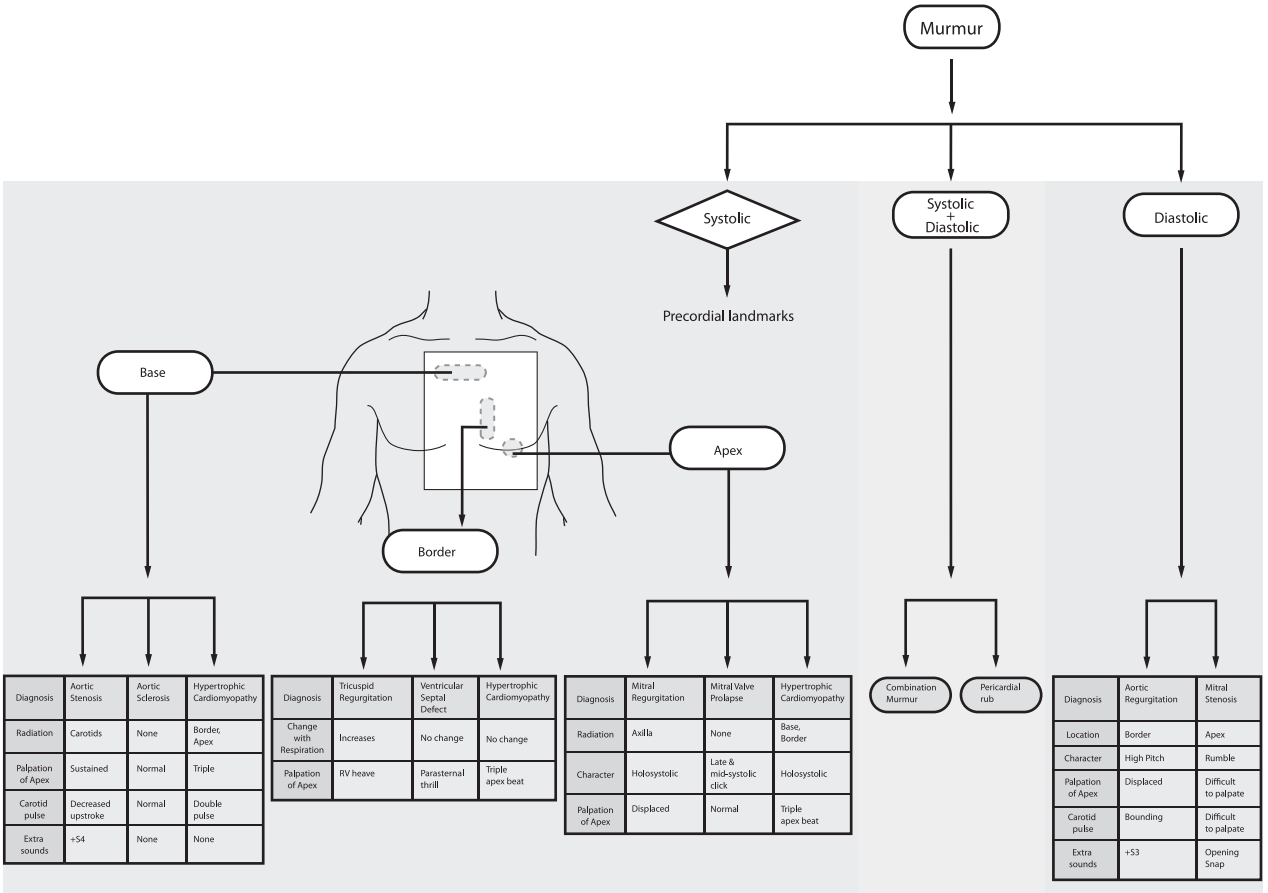


**Figure 2** Schema-based instructional framework

diagnosis. Two of the tested lesions were taught lesions from the learning phase (aortic regurgitation and mitral regurgitation). The remaining two lesions, aortic sclerosis and hypertrophic cardiomyopathy, were included on the instructional frameworks, but not in the learning phase (untaught lesions).

### Data analysis

An analysis of variance (ANOVA) was performed to compare written test scores. Unpaired *t*-tests were used to compare practical test scores and key feature identification. All analyses were performed with SPSS Version 17 (SPSS, Inc., Chicago, IL, USA).

The study was approved by the Research Ethics Board at the University of Toronto.

## RESULTS

A total of 53 students participated in the study; 26 were assigned to the schema-based and 27 to the traditional-based instruction.

### Learning phase

There was no difference in the time used to complete the learning phase between participants assigned, respectively, to the schema-based and traditional instructional frameworks (45 ± 11 minutes and 39 ± 8 minutes, respectively; $t_{12} = 1.2$, $p = 0.24$). Accuracy in diagnosing the four lesions in the learning phase was similar between those assigned to the schema-based and traditional frameworks (98% versus 93%, respectively; $t_{163} = 1.9$, $p = 0.06$). Students assigned to the traditional framework identified more total features (4.4 ± 1.8 versus 3.8 ± 1.1; $t_{210} = 2.9$, $p = 0.004$), more correct features not in the schema (1.4 ± 1.3 versus 0.1 ± 0.4; $t_{210} = 9.9$, $p < 0.001$), and more incorrect features (0.2 ± 0.6 versus 0.04 ± 0.2; $t_{130} = 3.1$, $p = 0.003$) compared with those assigned to the schema-based framework. Students who received the schema identified significantly more features present in the schema (3.7 ± 1.1 versus 3.0 ± 1.3; $t_{210} = 4.3$, $p < 0.001$).

*Table 1 Examples of questions on the written test*

| Structured knowledge questions* | Factual knowledge questions† |
|---|---|
| 1 You hear a diastolic murmur. Which features are most helpful in diagnosing the cause? Choose as many as you feel would be helpful | 1 Circle the features you would expect to find in a patient with aortic regurgitation. Choose as many as are correct |
| (a) JVP | (a) Bounding carotid pulse |
| (b) Grade of murmur | (b) S3 |
| (c) Location of murmur | (c) Normal apex |
| (d) Radiation of murmur | (d) Diastolic component of murmur |
| (e) Character | (e) Opening snap |
| (f) Apex beat | (f) CV waveform in JVP |
| (g) Carotid pulse | (g) Loudest at sternal border |
| (h) Changes with inspiration | (h) Changes with inspiration |
| 2 You hear a murmur loudest at the base. What physical examination manoeuvre would you do next to diagnose the murmur? Choose **one** response | 2 Which feature are you most likely to find in a patient with aortic stenosis? |
| (a) Examine the JVP | (a) Elevated JVP |
| (b) Examine the carotid pulse | (b) Murmur loudest at the sternal border |
| (c) Ask the patient to take a deep breath in and determine if the murmur changes | (c) Radiation to the carotids |
| | (d) S3 |

* Correct answers for structured knowledge sample questions: (1) c, d, e, f, g; (2) b
† Correct answers for factual knowledge sample questions:
(1) a, b, d, g; (2) c
JVP = jugular venous pressure

### Testing phase

Written tests were scored independently by two raters with high inter-rater reliability (Spearman's $r = 0.961$, $p < 0.001$). There was a significant effect of instruction type on performance ($F_{1003,3} = 17$, $p < 0.001$). There was also a significant interaction between instruction type and question type ($F_{1003,3} = 18$, $p < 0.001$). Schema-based instruction was associated with higher scores on structured knowledge questions (74% versus 55%; $p < 0.0001$), but not on factual knowledge questions (62% versus 62%; $p = 0.93$). On follow-up testing 2–4 weeks later ($n = 37$, 70% of participants), these findings remained unchanged. There was a significant effect of instruction type on performance ($F_{509,3} = 12$, $p = 0.001$) and a significant interaction between instruction type and question type ($F_{509,3} = 5$, $p = 0.02$). Students in the schema group performed better on structured knowledge questions (62% versus 42%; $p = 0.001$), but not on factual knowledge questions (52% versus 48%; $p = 0.41$).

On the practical test, schema-based instruction was associated with greater diagnostic accuracy (61% versus 26%; mean difference = 35%, 95% confidence interval [CI] 22–47; $t_{205} = 5.40$, $p < 0.001$). Schema-based instruction was associated with greater diagnostic accuracy for both taught (77% versus 39%, mean difference = 38%, 95% CI 20–56; $t_{102} = 2.26$, $p < 0.001$) and untaught (44% versus 13%, mean difference = 31%, 95% CI 15–48; $t_{89} = 3.75$, $p < 0.001$) lesions. For taught lesions, schema-instructed students identified the same number of overall features, more correct features in the schema, fewer correct features not in the schema and fewer incorrect features (Table 2). For untaught diagnoses, schema-instructed students identified fewer overall features, fewer correct features not in the schema and fewer incorrect features (Table 2).

## DISCUSSION

To our knowledge, this study is the first prospective, randomised study to evaluate schema-based

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*Table 2 Feature identification among students taught using the schema-based and traditional instruction frameworks*

| | All lesions | | Taught lesions | | Untaught lesions | |
|---|---|---|---|---|---|---|
| | **Mean difference (95% CI)*** | **p-value** | **Mean difference (95% CI)*** | **p-value** | **Mean difference (95% CI)*** | **p-value** |
| Correct features in schema | 0.39 (0.05–0.69) | 0.012 | 0.62 (0.23–1.02) | 0.002 | 0.15 (– 0.28 to 0.58) | NS |
| Correct features not in schema | – 0.71 (– 0.90 to – 0.51) | < 0.001 | – 0.64 (– 0.93 to – 0.35) | < 0.001 | – 0.77 (– 0.10 to – 0.51) | < 0.001 |
| Incorrect features | – 0.52 (– 7.32 to – 0.31) | < 0.001 | – 0.55 (– 0.80 to – 0.29) | < 0.001 | – 0.49 (– 0.83 to – 0.16) | 0.004 |
| Total features | 0.310 (– 0.64 to 0.049) | NS | 0.020 (– 0.52 to 0.48) | NS | – 0.62 (– 1.16 to 0.080) | 0.025 |

* Positive mean differences indicate that a greater number of features were identified by the schema-based instruction group; negative mean differences indicate that a greater number of features were identified by the traditional instruction group
NS = not significant

instruction among pre-clerkship learners. Our results provide strong support for the use of schema-based instruction in cardiac auscultation among novice students. We found that students instructed with a schema achieved higher diagnostic success than students instructed with the traditional framework. Of note, the 26% diagnostic accuracy achieved by the control group was comparable with rates achieved in previous evaluations of diagnostic accuracy among Year 2 medical students (19–24%),[12,19] which suggests that this group was not unfairly disadvantaged by our design.

How do schemas benefit novices? The hypothesis that schemas benefit novices by modifying knowledge organisation and diagnostic reasoning strategies is supported by our data. The results on the written test provide evidence that schemas modified knowledge organisation by increasing the amount of structured knowledge without changing factual knowledge of diagnoses. The magnitude of the effect (38–40%) is similar to that in a previous study in which schemas were found to improve scores on nephrology questions by 30%.[8] Although reasoning strategy was not directly measured (for fear of overwhelming novices with a verbal protocol), schema-instructed students were more likely to report features from the schema in arriving at their diagnosis. This suggests a higher use of schema-based reasoning by schema-instructed students.

Is there a downside to schema-based instruction? We anticipated lower diagnostic accuracy on untaught lesions among students instructed with schemas because they may be less likely to generate non-schema-based correct features. Although the traditional instruction group did generate a greater number of correct features for untaught lesions, this did not translate into higher diagnostic success. In fact, the schema-based instruction group had 30% higher diagnostic success. Two explanations are possible. Firstly, the traditional instruction group identified more incorrect features. Incorrect features may have a derailing effect on diagnostic reasoning. Secondly, the schema-based instruction group identified more schema-based features. Schema-based features may be more helpful in the diagnostic process than non-schema-based features as they help distinguish diagnoses. Importantly, we cannot generalise these findings to all diagnoses as we did not assess diagnoses or feature combinations not provided in the instructional material.

Our study has several limitations. Firstly, it was conducted at a single centre. Our conclusions would be strengthened by replication elsewhere. Secondly, practical testing was immediate. With further time to study and digest material, the benefit of a schema might be reduced. Thirdly, only novice learners participated. It is unclear how our findings might apply to experienced learners, who may already possess compiled knowledge. Fourthly, Harvey® has prominent and standardised physical examination findings that are not always present in real patients. This may have allowed students to make the correct

diagnosis with fewer variables, thereby favouring the schema group. Finally, not all complex clinical problems are easily translated into schemas. Schema development requires the identification of a small number of useful clinical variables, which may not always be apparent.

In summary, this study provides important proof-of-concept for a schema-based approach to clinical instruction. Like the traditional framework used in this study, current teaching methods often emphasise comprehensiveness and diagnosis-specific knowledge. Our randomised study suggests a substantial benefit in organising this knowledge using schemas. Future work should explore whether this benefit is found across other domains, in more advanced learners, and continues to persist over time.

*Contributors:* all authors contributed to the study design. SB collected the data. MS conducted the statistical analysis. All authors participated in data interpretation. SB drafted the manuscript and MS and RBC contributed to its critical revision. All authors approved the final manuscript for publication.

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